

One Ocean, Many Tasks: A Data-Grounded Simulation Suite and Benchmark for Marine Robotics

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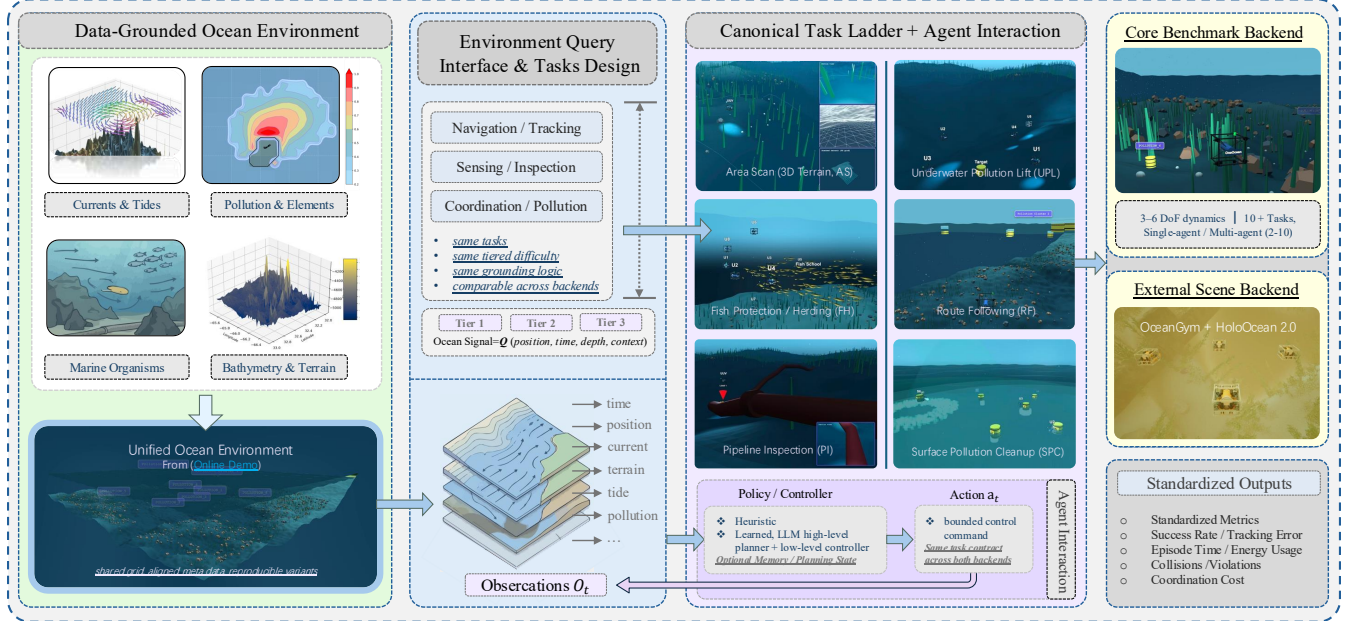


Figure 1: OneOcean overview. A unified ocean-environment product grounds a canonical multi-task benchmark and two complementary execution backends, enabling reproducible evaluation across navigation, sensing, coordination, and pollution-related marine-robotics tasks.

Abstract

Simulation is essential for marine robotics, yet existing platforms often trade off between visually realistic worlds, high-throughput experimentation, and ocean-condition realism grounded in real data. We present OneOcean, a data-grounded simulation suite and benchmark built around a unified spatiotemporal ocean-environment product that harmonizes bathymetry with data-driven currents and tides, with optional pollution fields for cleanup-related evaluation. On top of this environment representation, OneOcean defines a task ladder spanning navigation and station-keeping under currents, waypoint and route following, depth-profile tracking, area scanning, pipeline inspection, and multi-agent coordination, including formation transit, fish protection and patrol (herding), and surface pollution localization, containment, and cleanup. The suite supports multi-agent settings with 2–10 vehicles. Our simulator supports three to six degrees of freedom (DoF) and produces standardized metrics and reproducible run manifests, enabling consistent aggregation across tasks, difficulties, and scaling settings. Experiments across dataset variants and scenes systematically stress planning and control under data-grounded disturbances and reveal trade-offs among heuristic baselines, behavior cloning (BC), and a large language model (LLM)-based planner.

[Project](#) [Code](#) [Demo](#) [Hugging Face](#)

Introduction

Marine robots must operate under complex, highly variable ocean conditions. In practice, **currents and tides** can dominate vehicle motion, while field trials remain expensive and time-limited and at-sea sensing and communication are constrained (Zereik et al. 2018). Simulation is therefore essential, but existing platforms often trade off between (i) visually realistic underwater worlds, (ii) high-throughput, reproducible experimentation, and (iii) **ocean-condition realism grounded in real data**.

We present **OneOcean**, a data-grounded simulation suite and benchmark for marine robotics. OneOcean is built around a unified spatiotemporal ocean-environment dataset in Network Common Data Form (NetCDF). It harmonizes bathymetry from the General Bathymetric Chart of the Oceans (GEBCO) (GEBCO Compilation Group 2025) with data-driven currents and tides from the Copernicus Marine Service Global Ocean Physics Analysis and Forecast (GOPAF) product (Copernicus Marine Service 2024; Copernicus Marine Environment Monitoring Service (CMEMS) 2025). On top of this environment representation, OneOcean defines a **10-task ladder** spanning single-agent navigation and tracking, spatial sensing and coverage, and multi-agent coordination and pollution-related tasks, with support for teams of 2–10 vehicles. The same environment product is also packaged into multiple dataset variants so

the benchmark can support fast local iteration, scene-oriented construction, and broader public release without changing task semantics. Figure 1 summarizes this organization.

OneOcean provides two complementary execution backends that share task definitions and evaluation semantics. The *core benchmark backend* is deterministic and throughput-first, producing standardized metrics and reproducible run manifests for consistent aggregation. The *external underwater-scene backend* reuses packaged scenes from an existing simulator stack and injects OneOcean’s data-grounded currents, enabling qualitative evaluation in visually realistic underwater settings while preserving the benchmark protocol. This split separates controlled large-scale benchmarking from scene-rich portability studies while grounding both backends in the same ocean data product. In summary, our contributions are:

- A unified, reusable ocean-environment data product and tooling for exporting data-grounded disturbance signals to simulators, with variants that support fast iteration, scene construction, and broader public release.
- A canonical 10-task benchmark suite with difficulty levels, standardized metrics, auditable run manifests, and support for 3–6 DoF vehicle dynamics, with 6DoF used for the main quantitative evaluation.
- Experiments across the two backends that reveal task difficulty, scaling, and planner-compute trade-offs under data-grounded disturbances.

Related Work

Classical simulators built on Gazebo (Koenig and Howard 2004), including UUV Simulator (Manhães et al. 2016) and UWSim (Prats et al. 2012), provide dynamics and sensor stacks commonly used in robotics workflows, but they are not designed for data-grounded ocean-condition benchmarking at scale. More recent systems target higher-fidelity rendering or higher-throughput experimentation, including MARUS (Lončar et al. 2022), OceanSim on Isaac Sim (Song et al. 2025; Makovychuk et al. 2021), DAVE (Zhang et al. 2022), MarineGym (Chu et al. 2025), and URLab, which integrates Unreal Engine with MuJoCo (Embley-Riches et al. 2026). These platforms provide important simulator infrastructure, but they typically do not expose a reusable ocean-condition product, a fixed task ladder, and a reporting protocol that remain consistent across multiple execution backends. Reproducing the same ocean-conditioned task across simulators therefore requires substantial custom engineering. Table 1 summarizes this distinction at both the system and task levels.

Ocean-condition realism is commonly provided by numerical models and operational forecasting systems. Representative community models include Delft3D (Lesser et al. 2004), NEMO (Madec and the NEMO Team 2019), MITgcm (Strobach et al. 2018), ROMS (Shchepetkin and McWilliams 2005), and HYCOM (HYCOM Consortium for Data-Assimilative Ocean Modeling 2024). Operational products from the Copernicus Marine Service, including GOPAF, provide data-assimilative currents and hydrographic variables.

GEBCO provides global gridded seafloor elevation, while NetCDF stores labeled multidimensional coordinates, variables, units, and provenance in a self-describing representation. These sources are public, globally scoped, and machine-queryable, allowing the same ocean condition to be extracted and audited across runs. Toolkits such as Parcels (Delandmeter and van Sebille 2019) and OpenDrift (Dagestad, Röhrs et al. 2018) support Lagrangian drift and dispersion analysis. OneOcean complements these efforts by treating ocean conditions as a reusable spatiotemporal product for robotics and pairing them with a protocol that runs across multiple simulator backends.

Recent task-specific marine systems provide complementary evidence. DREAM studies vision-language monitoring through simulation and tank trials (Wu et al. 2026); Map-TIDAL allocates cooperative underwater-inspection tasks (Ruediger and Banerjee 2026); and Douro MARL maps a data-driven river plume with multiple vehicles (Dal Fabbro et al. 2026). Recent AI benchmarks broaden the evaluation perspective: CAMAR defines high-throughput continuous multi-agent routing and a three-tier protocol (Pshenitsyn, Panov, and Skrynnik 2026); RobotArena ∞ converts real demonstrations into scalable simulation tests for robot policies (Jangir et al. 2026); and CaP-X evaluates executable embodied agents across multiple manipulation suites (Fu et al. 2026). OneOcean addresses the remaining marine-specific gap by combining standardized environmental conditions, a canonical task suite, auditable run manifests, and shared contracts across two execution backends.

OneOcean Suite

OneOcean is organized around a single principle: treat *ocean conditions* as a reusable, auditable product, and evaluate robotics methods on a canonical task ladder under these conditions. This section summarizes the environment product, how it is consumed by the benchmark backends, and the task and evaluation definitions.

Unified spatiotemporal ocean-environment product

The shared NetCDF representation harmonizes GEBCO bathymetry with GOPAF currents and optional tides. It exposes named latitude, longitude, time, and depth coordinates, eastward and northward current components, elevation, and a mask for invalid terrain and NoData cells. The pipeline crops the requested region, retains the returned time and depth levels, aligns variables on one horizontal grid, and records provenance.

The core backend selects a fixed (t, d) slice for deterministic rollouts, whereas the external backend advances a dataset clock. Table 2 reports the realized products: the 0–200 m requests returned 26 levels through 186.13 m, while *tiny* contains one near-surface level at 0.494 m. Appendix Sec. A specifies the full schema, mask semantics, coordinate conventions, validation results, and tide and depth caveats.

The mask denotes invalid terrain, crop-exterior, or NoData cells rather than a guaranteed coastline classification, and consumers must not interpret zero-filled masked elevation as seabed. Likewise, the single *tiny* layer is not evidence of resolved 3D flow. In the tide-enabled scene product, higher-

Symbols: ● Provided; ● Not provided. Tasks: G2G, SK, RF, DPT, AS, PI, FT, SPC, UPL, and FH.

Platform or Work	Venue	Real-data In Loop	Multi- agent	Cross- backend	3D Currents	Pollution or Spill	Tide Modeling	Canonical Task	Metrics Protocol	Headless Batch	G2G	SK	RF	DPT	AS	PI	FT	SPC	UPL	FH
MIMIR-UW	2023	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
HoloOcean 2.0	2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
OceanSim	2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
MarineGym	IROS 2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
OceanGym	2025	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
URLab	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
DREAM	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Douro MARL	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Map-TIDAL	ICRA 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
CAMAR	AAAI 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
RobotArena ∞	ICLR 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
CaP-X	ICML 2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
OneOcean (ours)	2026	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●

Table 1: System-level comparison and task coverage. Entries are conservatively coded from the cited papers and their public artifacts (Álvarez-Tuñón et al. 2023; Romrell et al. 2025; Song et al. 2025; Chu et al. 2025; Xue et al. 2025; Embley-Riches et al. 2026; Wu et al. 2026; Dal Fabbro et al. 2026; Ruediger and Banerjee 2026; Pshenitsyn, Panov, and Skrynnik 2026; Jangir et al. 2026; Fu et al. 2026).

Variant	Region and period	$T \times D \times H \times W$	Resolution; size; use
tiny	42.1–42.7 N, 71.2–70.2 W; 2025	$357 \times 1 \times 62 \times 101$	$\approx 0.01^\circ$; 29 MB; tests and web
scene	32–33 N, 66.5–65.5 W; Dec. 2025	$23 \times 26 \times 240 \times 240$	0.00418° ; 690 MB; scenes
public	30–40 N, 72–62 W; 2025	$357 \times 26 \times 41 \times 41$	0.25° ; 175 MB; release

Table 2: Realized combined NetCDF variants. Sizes are the merged products themselves; source subsets and derived assets are distributed separately.

frequency tide components are aligned to the base clock and attenuated with depth; this documented engineering profile supports controlled disturbance tests but not phase-accurate 3D tidal simulation.

Data-grounded disturbance signals

OneOcean exports either a fixed gridded slice for high-throughput rollouts or a temporal profile for scene-backed execution. Both apply current through a recorded relative-velocity convention in the core frame $(x, y, z) = (\text{east, depth-down, north})$, and every artifact records its extraction indices. Figure 2 visualizes one grounded current slice over bathymetry; Appendix Sec. A details the export and provenance contract.

Environment queries and probes. Shared queries expose local current, bathymetry, mask validity, and task signals. Pollution tasks observe $o_i^c(t) = C(p_i(t), t)$ from either a current-advected Gaussian or a regeneratable Ocean Current Prediction Network (OCPNet) field; these are controlled signals, not in-situ labels. Appendix Sec. C gives the probe, sink, and discrete-source cleanup semantics.

Task ladder and difficulty levels

OneOcean defines ten tasks spanning navigation and tracking (G2G, SK, RF, DPT), sensing and inspection (AS, PI), and coordination and pollution response (FT, SPC, UPL, FH). Table 3 gives team sizes and primary metrics; Appendix Sec. B provides the complete objective, observation, constraint, and success contracts.

Tiers 1, 2, and 3 jointly increase current strength, precision, route and coverage extent, sensing difficulty, and coordination

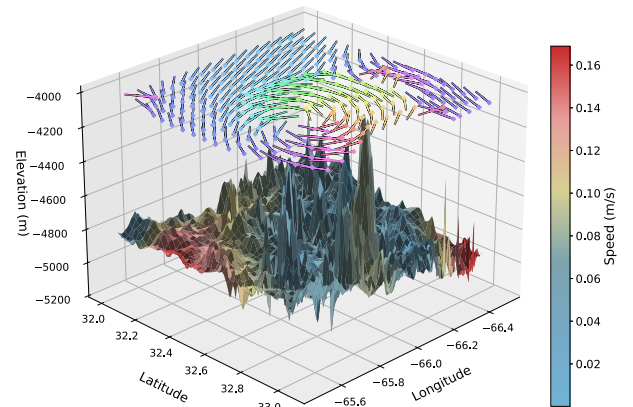


Figure 2: Base current field visualization over bathymetry. The horizontal current components are u_{data} and v_{data} ; surface color indicates speed magnitude (m s^{-1}). View configuration: pitch 25° , azimuth 45° , and depth index 0 m. The terrain surface is colored by speed and overlaid with arrows indicating current direction.

Category	Task	Agents	Goal	Primary metric
Navigation	G2G	1	reach	final goal error
Navigation	SK	1	hold	station root-mean-square error
Navigation	RF	1	follow	cross-track error
Navigation	DPT	1	track	depth-tracking error
Sensing and Coverage	AS	1–10	scan	coverage fraction
Sensing and Inspection	PI	1–10	inspect	inspection score
Coordination	FT	2–10	transit	formation error
Pollution	SPC	2–10	clean	cleanup rate
Coordination	UPL	5	lift	lift success
Coordination	FH	8	herd	containment success

Table 3: Canonical task ladder used in OneOcean (10 tasks). Each task is evaluated at Tiers 1–3 and paired with a compact goal type and a primary task-specific metric.

tightness while preserving each task’s interface. Every task declares its objective, observations, feasibility constraints, success condition, and primary metric. The release also includes a lightweight Robot Operating System 2 (ROS 2) bridge for state, probes, and velocity commands; its frame, topic, and service contract appears in Appendix Sec. C, while ROS performance remains outside this evaluation.

Navigation and tracking tasks test disturbance rejection and sustained trajectory accuracy; sensing and inspection tasks require spatial coverage or physical cue encounters; coordination and pollution tasks add team geometry, assignment, and dwell-time coupling. This grouping keeps the shared reporting contract compact without collapsing task-specific progress into success alone.

Two execution backends

OneOcean provides two complementary execution backends that share task and reporting contracts. The **core benchmark backend** is headless and throughput-first: it consumes a NumPy archive (NPZ) slice plus task and action configuration and returns observations, metrics, manifests, and recorder streams at scale. It supports 3–6 DoF engineering dynamics (6DoF in the main experiments) (Manhães et al. 2016). The **external underwater-scene backend** prioritizes scene reuse: it consumes a temporal current export and the same task semantics inside packaged underwater scenes, returning quantitative route and task outputs with curated media. Thus, the second backend studies portability and scene realization; it is not claimed as a new rendering engine or as an architecture-matched throughput comparison.

Benchmark protocol and metrics. Actions specify desired relative velocity, clipped as $a_t = \text{clip}(\tilde{a}_t; v_{\max})$. With current gain g ,

$$c_t = g[u_{\text{data}}(x_t, z_t), 0, v_{\text{data}}(x_t, z_t)] + c_{\text{tide}}(t), \quad (1)$$

where u_{data} and v_{data} are sampled horizontal currents and c_{tide} is an optional synthetic robustness signal. With body velocity $\nu = [\nu_\ell^\top, \nu_a^\top]^\top$, rotation R , and diagonal M, D, K_ℓ, K_a, K_d , the 6DoF engineering update is

$$\begin{aligned} M\dot{\nu} &= \tau - D\nu, & \tau_\ell &= K_\ell(\nu_\ell^{\text{cmd}} - \nu_\ell), \\ \tau_a &= K_a(\nu_a^{\text{cmd}} - \nu_a) - K_d\nu_a, & \dot{p} &= R\nu_\ell + c_t. \end{aligned} \quad (2)$$

Commands are transformed to the body frame; yaw follows horizontal motion while roll and pitch are stabilized. Kinematic mode directly integrates $a_t + c_t$, and 3DoF suppresses roll and pitch. The diagonal model is not vehicle-identified hydrodynamics: it omits added mass, Coriolis and restoring

terms, thruster allocation, and two-way fluid coupling. Hard terrain feasibility is

$$\mathcal{C} = \{(x, y, z) \mid m_{\text{land}}(x, z) < \tau_{\text{land}}, y + \delta_{\text{clr}} \leq -h(x, z)\}, \quad (3)$$

where m_{land} is mask validity, h is bathymetric elevation, and δ_{clr} is required clearance. Masked, out-of-bounds, or clearance-violating proposals are rejected through the shared constraint path. We report SR, timeout-aware episode time T , $E = \sum_t \|a_t\|_2^2 \Delta t$, near-collision ratio, and task-specific metrics.

BC maps normalized goal, depth, probe, current, and task features ϕ_t to $\hat{a}_t = f_\theta(\phi_t)$ and minimizes the heuristic-action loss

$$\mathcal{L}_{\text{BC}} = N_{\text{BC}}^{-1} \sum_t \|a_t^{\text{heur}} - \hat{a}_t\|_2^2. \quad (4)$$

Every K steps, an LLM instead proposes a discrete assignment $g_k = \pi_{\text{LLM}}(s_k)$; schema-invalid output invokes a deterministic allocator, and a shared low-level controller executes $a_t = \pi_{\text{low}}(x_t, g_k)$. Appendix Sec. C gives dynamics defaults, pollution semantics, and ROS mapping; Appendix Sec. D defines all metrics and manifests.

Optional pollution fields (OCPNet). Continuous containment applies a local concentration sink, whereas canonical SPC completes discrete sources after an assignment-and-dwell condition; its cleanup rate is therefore source completion, not removed fluid mass. Figure 3 shows the regeneratable multi-species signal.

Experiments

We evaluate OneOcean along four axes: method separation on the unified 10-task suite under 6DoF settings; difficulty and control-versus-planning gaps across the tiered task ladder; trade-offs in success, efficiency, and compute among heuristic control, BC, and local LLM planners; and transfer of the same task semantics to the external underwater-scene backend grounded by the same ocean data product.

Experimental setup

The main core results use 6DoF with hard mask and clearance checks and a fixed NetCDF-derived current slice; the external backend advances a temporal export in four packaged scenes. We report SR, timeout-aware T , action-energy proxy E , near-collision ratio, and the task metrics named in the tables. Headline budgets are stated with their result tables, while separate diagnostics audit planner decisions, dynamics sensitivity, and pollution observations. All runs store their integer seed, task and dynamics configuration, dataset indices, and software and Git state. Local LLM inference assigns one 48 GB accelerator to each process. Appendix Sec. D specifies the supplementary diagnostic budgets and complete manifest contract.

Here T uses success time on successful episodes and timeout on failures; E is computed only from clipped vehicle commands and is not inference energy. Coll(%) counts steps in which the closest agent pair is within the configured near-collision radius.

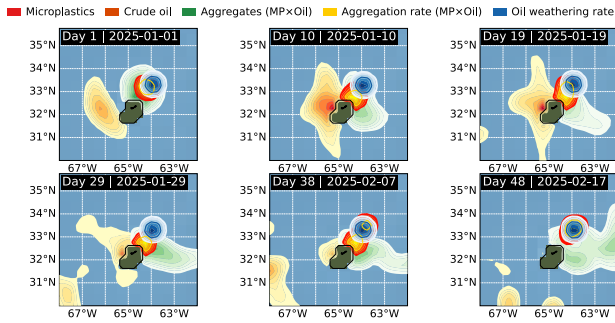


Figure 3: Curated OCPNet multi-species diffusion and reaction field used as qualitative evidence. We treat these fields as regeneratable signals that can be coupled to localization, containment, and cleanup tasks through probes and task-specific metrics.

Method	Macro (Tier 2)			Macro (Tier 3)			Tier 3 key-task metrics					
	Suite SR(%)	E	Coll. (%)	Suite SR(%)	E	Coll. (%)	G2G SR(%)	RF SR(%)	SK RMS	Scan cov.	Pipe score	Cleanup rate
Heuristic	84.6	1842.3	5.68	29.0	3352.4	8.19	74.0	54.0	1.07	0.89	0.78	0.56
BC	58.6	1271.2	9.72	6.0	1635.1	6.41	34.0	0.0	26.52	0.89	0.67	0.06

Table 4: Headline core-benchmark results (6DoF): macro averages at Tiers 2 and 3 plus Tier 3 key-task metrics, using 50 episodes per task and tier. Success rate (SR) is reported in percent; Macro Suite SR is an unweighted mean across the 10 canonical tasks. Coll is the mean collision-step ratio (%) with a 2.0 m radius. Best values per column are highlighted.

Method	Avg. SR(%)	Cleanup SR(%)	Cleanup rate	Scan SR(%)	Scan cov.	Pipe SR(%)	Pipe score	$T(s)$	E
Heuristic	65.8	60.0	0.87	77.5	0.80	60.0	0.69	776.8	9132.4
BC	45.0	10.0	0.32	67.5	0.79	57.5	0.61	921.6	6908.2
LLM-ChatGLM3-6B	55.0	60.0	0.87	77.5	0.82	27.5	0.54	860.3	9671.5
LLM-Qwen2.5-7B	47.5	57.5	0.81	57.5	0.78	27.5	0.57	1077.2	11703.8
LLM-Qwen2-7B	45.0	57.5	0.86	67.5	0.81	10.0	0.51	991.4	10422.1
LLM-Mistral-7B	42.5	55.0	0.82	55.0	0.78	17.5	0.52	1044.2	11288.9
LLM-Llama3-8B	44.2	67.5	0.84	47.5	0.77	17.5	0.59	1099.1	11614.2
LLM-Llama2-7B	30.8	55.0	0.87	27.5	0.75	10.0	0.46	1156.4	11654.3

Table 5: Core-benchmark planning-suite performance at Tier 2 with doubled currents: heuristic vs BC vs local LLM planners on three planning-sensitive tasks, using 40 episodes per task (120 per method). Best values per column are highlighted.

Task	N	Heur T2	BC T2	Heur T3	BC T3
<i>Navigation and Tracking</i>					
G2G	1	94.0	44.0	74.0	34.0
SK	1	100.0	0.0	14.0	0.0
RF	1	84.0	24.0	54.0	0.0
DPT	1	84.0	64.0	50.0	0.0
<i>Coordination</i>					
FT	10	100.0	40.0	20.0	0.0
FH	8	94.0	100.0	4.0	4.0
<i>Sensing and Pollution</i>					
PI	1	100.0	90.0	30.0	14.0
AS	1	90.0	84.0	14.0	4.0
SPC	10	84.0	50.0	10.0	0.0
UPL	5	16.0	90.0	20.0	4.0

Table 6: Per-task success rates at Tier 2 and Tier 3 for the canonical 10-task suite, using 50 episodes per task and tier. Highlighted rows are revisited in the planning and external-backend transfer studies.

Method	Avg. SR(%)	Calls	Lat. (s)	ms per call	Prompt tok.	Output tok.	Total tok.
LLM-Qwen2.5-7B	65.0	6.48	14.21	2222.1	3932	4084	8016
LLM-Qwen2-7B	53.3	4.71	11.24	2440.3	2877	2996	5873
LLM-Mistral-7B	53.3	6.05	31.38	4942.5	3762	4128	7890
LLM-ChatGLM3-6B	53.3	5.38	7.91	1393.6	3414	411	3825
Heuristic	55.0	0.00	0.00	0.0	0	0	0
LLM-Llama3-8B	45.0	6.71	13.94	2117.5	3072	3218	6292
LLM-Llama2-7B	33.3	6.59	49.42	7401.8	4219	4851	9071
BC	21.7	0.00	0.00	0.0	0	0	0

Table 7: Core-benchmark planning-suite cost metrics at Tier 2 with doubled currents: uncached local-inference calls, latency, and token usage, reported alongside average success over 20 episodes per task.

Coverage, inspection, source completion, station error, cross-track error (CTE), and reached waypoints expose progress that binary SR would hide. Tiers 1–3 jointly change disturbance, tolerance, spatial extent, and coordination settings rather than varying one scalar.

Core benchmark backend

Figure 4 shows four representative tasks from the canonical suite. The quantitative evidence in Tables 4 and 6 establishes two main properties of the benchmark. First, Tier 2 is chal-

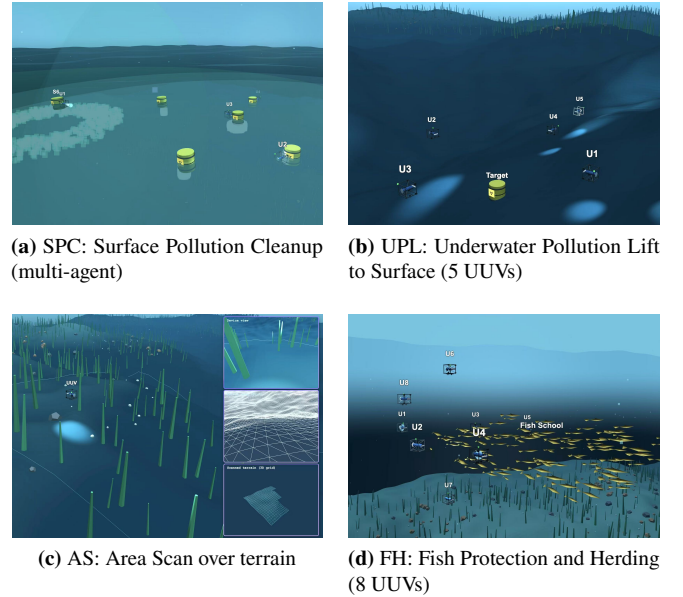


Figure 4: Representative core-benchmark tasks used in this work’s experiments. These snapshots connect the quantitative tables to the task geometry and multi-agent interactions used in the benchmark.

lenging but not saturated for all methods, making it useful for regression testing and planner comparison. Second, Tier 3 exposes large method gaps on both single-agent and multi-agent tasks, especially in station keeping, route following, surface cleanup, and leak inspection.

Figure 5 makes the ladder effect explicit on a representative task subset: the left panel shows the tier-wise success profile, while the right panel isolates where the largest drops occur as the benchmark moves from Tier 1 to Tier 3.

Two observations stand out. Heuristic control remains the strongest general-purpose baseline in the core benchmark backend, especially once Tier 3 activates stricter tolerances and stronger drift. BC remains competitive on *Underwater-Lift* and *Fish-Herding* at Tier 2. However, it degrades sharply on

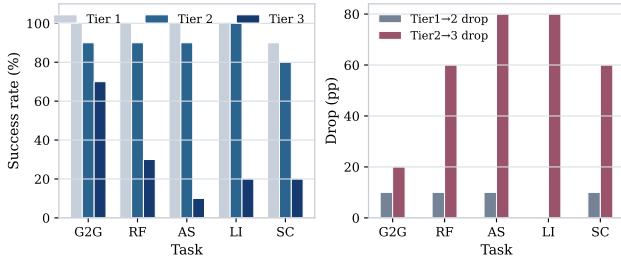


Figure 5: Difficulty-ladder summary on a representative five-task subset. Left: success rate at Tiers 1–3. Right: the corresponding success-rate drops from Tier 1→2 and Tier 2→3, highlighting that the hardest setting remains strongly discriminative for route following, area scan, leak inspection, and surface cleanup.

tasks that require persistent disturbance rejection or coordinated geometry maintenance. Figure 6 further shows that the surface-cleanup task captures a meaningful team-size trade-off: larger teams improve success and source-completion rate, but collision risk increases and the energy budget does not scale monotonically.

Planning-sensitive tasks. We next focus on three planning-sensitive tasks: surface cleanup, area scan, and leak inspection. We compare deterministic heuristic control, BC, and several local LLM planners. The LLM never replaces low-level control; it proposes high-level assignments or waypoint choices at a fixed stride, while the low-level controller and safety checks remain deterministic.

Tables 5 and 7 show a useful separation. The heuristic remains the most reliable overall planner, but several local LLMs are competitive on specific tasks: ChatGLM3-6B matches the heuristic on cleanup rate and scan coverage with lower latency than the other LLMs, while Qwen2.5-7B gives the best average success among the LLM family in this sweep.

An instrumented three-seed audit explains why valid language output does not imply task success: fallback can invoke a competent allocator, while valid plans can still repeat covered regions or miss leaks. Thus E measures executed trajectories, whereas latency and token counts measure inference cost. The LLM rows are high-level planning diagnostics rather than architecture-matched low-level controller rankings. Appendix Sec. E reports task-level and model-level validity, fallback, completion, token, and latency values.

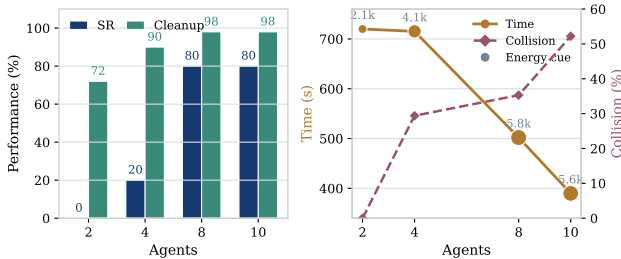


Figure 6: Surface-cleanup scaling at Tier 2. Left: success rate and cleanup rate as the number of vehicles increases. Right: completion time and near-collision ratio, with per-point annotations showing the energy proxy in thousands. The figure highlights that scaling improves cleanup outcomes, but also shifts the coordination cost profile.

Scene	Tier 2			Tier 3		
	SR (%)	WP	CTE (m)	SR (%)	WP	CTE (m)
Dam	0.0	1.0	3.06	0.0	0.0	0.87
OpenWater	20.0	3.1	2.50	0.0	0.0	1.91
PierHarbor	56.7	3.5	2.22	0.0	2.0	1.41
SimpleUnderw.	100.0	4.0	3.04	96.7	3.0	2.16
Mean over scenes	44.2	2.92	2.70	24.2	1.25	1.59

Table 8: Route-following across four underwater scenes after the success-criterion fix. Each scene uses 30 episodes per difficulty. “Waypoints” is the mean number of reached waypoints and CTE is cross-track error.

External underwater-scene backend



(a) Surface pollution containment and cleanup in a packaged ocean scene. **(b)** Multi-agent scene-backed deployment.

Figure 7: Qualitative evidence from the external underwater-scene backend, which reuses packaged underwater worlds while injecting currents exported from the same combined dataset used by the core benchmark backend.

Figure 7 and Table 8 show that the shared task contract remains executable but route following degrades differently across scenes. A separate diagnostic found lower localization error with OCPNet-3D under both tested current settings without reducing aggregate success. Appendix Sec. F provides the values, episode budget, and interpretation limits.

The external underwater-scene results should be read as a portability study rather than as a replacement for core-benchmark throughput evaluation. This backend reuses realistic scenes and produces paper and web media, but it also introduces stricter scene geometry and a different disturbance hook. Both backends retain the same task semantics, data provenance, and reporting contract.

Discussion and Limitations

OneOcean’s data-grounded disturbances provide repeatable stress tests across task families. Tier 2 supports regression testing, whereas Tier 3 exposes failures in tracking and coordination. No method dominates every criterion: heuristic control is strongest overall, BC remains competitive on selected structured tasks, and local LLM planners trade improved planning behavior for additional latency and token cost.

Success alone can obscure substantial partial progress and differences in energy, collision exposure, or completion time. We therefore pair success with task-specific progress. Planner validity is reported separately from closed-loop performance because a syntactically valid assignment may be dynamically poor, while deterministic fallback can conceal a failed planner call. Backend portability is assessed through shared task contracts, data provenance, output semantics, and common con-

straint checks rather than direct visual-fidelity or throughput rankings between unlike simulators. A controller, vehicle, or backend is compatible only when it consumes the environment query contract, declares termination conditions and metrics, records its configuration and dataset indices, and preserves the shared constraint path.

The validator and ROS 2 smoke checks establish interface compliance rather than physical fidelity or task performance. OneOcean remains a controlled engineering abstraction rather than a full hydrodynamic simulator. Its diagonal 6DoF update omits added mass, Coriolis and restoring forces, thruster allocation, and two-way fluid coupling. Dataset currents use a recorded relative-transport convention; the external backend applies them through a one-way disturbance hook. The `tiny` product has one near-surface layer, whereas the larger products retain 26 measured horizontal-current levels. OCPNet nevertheless repeats horizontal current over model depth and assumes zero vertical current. Appendix Sec. D details the remaining pollution, tide, ROS 2, and extension boundaries.

Conclusion

OneOcean combines a unified spatiotemporal ocean-environment product, a canonical 10-task ladder, and standardized evaluation for marine robotics. Its headless and underwater-scene backends support reproducible experiments under data-grounded disturbances. Comparisons of heuristic control, BC, and local LLM planners demonstrate the suite’s utility for quantitative evaluation, robustness and planner-cost analysis, and scene-backed portability studies. Future work will improve tide coupling and three-dimensional currents, broaden tasks and robot morphologies, and extend the release of derived datasets and benchmark artifacts.

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Appendix

This appendix details the environment representation, task objectives, motion and pollution models, and evaluation protocol. It also reports complementary studies of dynamics, disturbance strength, planner decisions, and scene-backed multi-agent behavior.

A Environment Product Contract and Validation

Sources, harmonization, and schema

OneOcean combines bathymetry from the General Bathymetric Chart of the Oceans (GEBCO) (GEBCO Compilation Group 2025) with horizontal currents from the Copernicus Marine Service Global Ocean Physics Analysis and Forecast (GOPAF) product (Copernicus Marine Service 2024; Copernicus Marine Environment Monitoring Service (CMEMS) 2025). The resulting Network Common Data Form (NetCDF) product is harmonized by cropping the requested region, retaining returned time and depth coordinates, normalizing coordinate names and units, interpolating variables onto one latitude–longitude grid, preserving invalid terrain and NoData cells, and recording the transformations. Harmonization does not mean inventing unavailable depth layers.

The storage merge fills masked elevation cells with zero, so elevation must always be interpreted together with the validity mask. A zero at an invalid cell is not a physical seabed measurement. In the core simulation frame, $(x, y, z) = (\text{east}, \text{depth-down}, \text{north})$, and the dataset vector (u, v) maps to $(u, 0, v)$. Longitude and latitude are converted to local metric x and z about the recorded lower-left geographic origin.

Depth and tide treatment

The realized `tiny` request contains one level at 0.494 m. The scene and public requests cover 0–200 m and retain 26 source levels from 0.494 to 186.126 m. The pipeline records returned coordinates rather than relabeling a nearest source level as the requested value. Consequently, `tiny` is a time-varying 2D or 2.5D current product, whereas the other two products provide measured horizontal current at multiple discrete depths.

Only the current scene release includes tide components. Hourly tide samples are aligned to the base product with nearest-time selection and attenuated as $\exp(-d/z_0)$ with $z_0 = 50$ m before forming the total current pair. This is a documented engineering approximation chosen for data volume and simulation convenience. It discards high-frequency structure on coarser time grids and is neither a validated 3D tidal model nor suitable for studies that require phase-accurate tidal dynamics.

Validator evidence

The release validator checks required fields, aliases, dimensions, monotonic coordinates, units, finite sampled values, mask semantics, provenance, and optional round-trip fidelity between NetCDF and a NumPy archive (NPZ). Table A2

summarizes the July 21, 2026 audit of the already-generated products.

The legacy warnings identify missing depth or elevation units, total-current units, or mask flag meanings depending on the variant. Regeneration with the updated pipeline writes these attributes. The audited `tiny` slice preserved coordinates and currents at absolute tolerance 10^{-6} .

Portable provenance

Every regenerated NetCDF records source product identifiers, requested and actual region, time, and depth, tide inclusion and depth profile, interpolation and missing-value handling, and generation software. Companion variant manifests use paths relative to the variant directory; private workspace paths are excluded. This contract allows a custom product to fail early before an episode is launched and lets every run retain the selected time and depth indices.

Geographic sampling within an episode

Let (ϕ_0, λ_0) be an episode’s geographic origin and $p = (x, y, z)$ its local position in meters. The core sampler uses

$$\phi(p) = \phi_0 + z/L, \quad (\text{A1})$$

$$\lambda(p) = \lambda_0 + x/[L \max(0.1, \cos \phi_0)], \quad (\text{A2})$$

where $L = 111132$ m per degree and the cosine evaluates latitude in radians. At reset, the origin is sampled within the available geographic extent to place the simulation tile on the environmental grid. The default tile spans 600×600 m horizontally, with vehicle depth bounded between 2 and 18 m. Environmental coordinates retain the source grid spacing.

For grid coordinates $\{\phi_j\}$ and $\{\lambda_k\}$, the sampled indices are

$$j^* = \arg \min_j |\phi_j - \phi(p)|, \quad k^* = \arg \min_k |\lambda_k - \lambda(p)|. \quad (\text{A3})$$

The current at p is $g(u_{j^*k^*}, 0, v_{j^*k^*})$, where g is the current gain. Bathymetry and validity use the same indices. Missing current samples are set to zero, while terrain feasibility is determined separately by the validity mask and seabed clearance. Core episodes query a fixed time and depth slice.

Temporal forcing in the scene backend

The scene backend uses an exported current sequence at a selected geographic location and the nearest available source depth. Let T_c be its number of temporal samples, s the episode seed, and α the configured sequence-advance rate. At simulation time t , the sampled index is

$$i(t) = [(s \bmod T_c) + \lfloor \alpha t \rfloor] \bmod T_c. \quad (\text{A4})$$

Setting $\alpha = 0$ holds the initial sample; a positive value advances the sequence and wraps at its end. For a daily export, α gives source days per simulated second. The sampled vector, multiplied by the current gain, drives the scene disturbance. Spatial slices and temporal sequences provide complementary conditions for evaluating disturbance rejection.

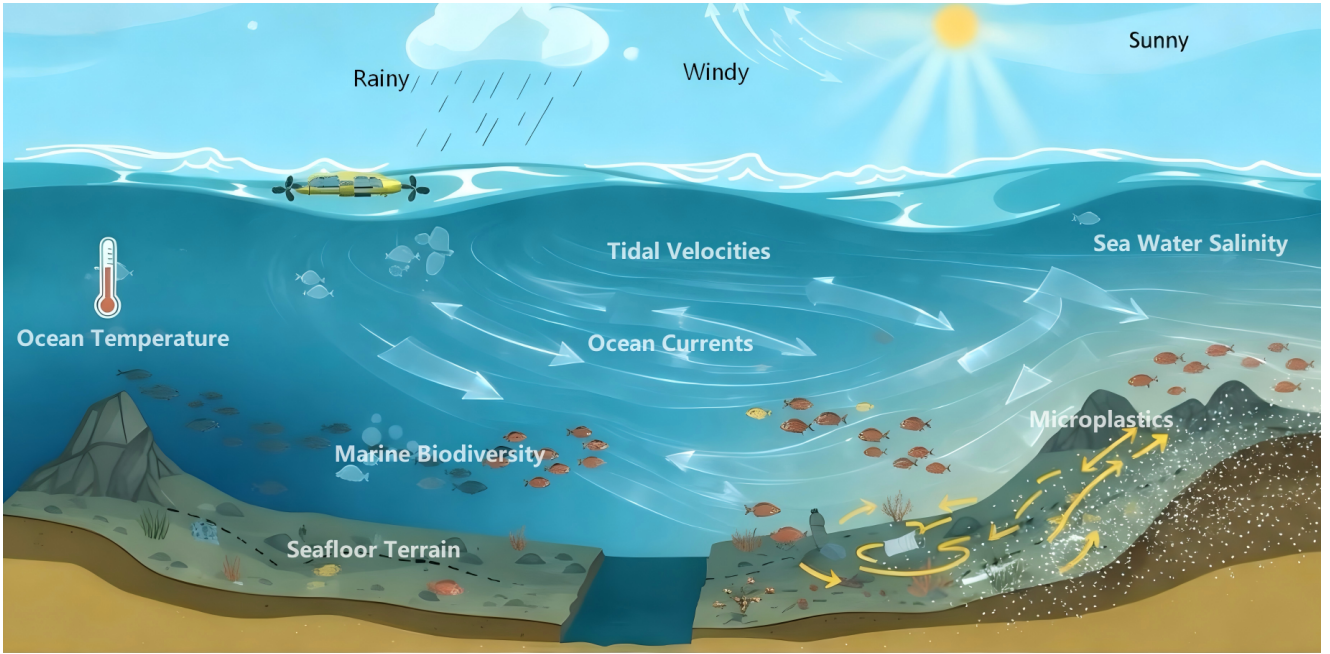


Figure A1: Conceptual view of the environmental channels and ecological signals represented by, or targeted for coupling with, OneOcean. It illustrates terrain, currents, tidal velocity, thermohaline variables, weather, biodiversity, and microplastics; it is not a scale diagram and does not imply that every release variant contains every depicted field.

Canonical field	Dimensions	Unit or Encoding	Contract
Latitude	H	degrees north	Strictly monotonic geographic coordinate.
Longitude	W	degrees east	Strictly monotonic geographic coordinate.
Time	T	Climate and Forecast time unit	Monotonic timestamps retained from the source product.
Depth	D	m, positive down	Actual source levels; a one-level product is valid but not a resolved 3D flow.
Eastward and northward current	$T \times D \times H \times W$	m s^{-1}	Base pair (u, v) or tide-combined pair $(u_{\text{total}}, v_{\text{total}})$.
Elevation	$H \times W$	m relative to sea surface	Bathymetric elevation aligned to the current grid.
Validity mask	$H \times W$	binary	Invalid terrain, crop-exterior, or NoData cells; not necessarily a coastline classifier.

Table A1: Canonical environment-product schema. The validator accepts documented aliases but maps every accepted product to these semantics.

Variant	Current pair	Errors	Warn.	Round-trip
tiny	uo, vo	0	3	Exact
scene	utotal, vttotal	0	4	Not run
public	uo, vo	0	2	Not run

Table A2: Environment-product validation. All variants are PASS_WITH_WARNINGS; warnings concern legacy metadata rather than numeric or schema errors. Round-trip denotes the fixed tiny ($t = 0, d = 0$) NetCDF-to-NPZ check.

B Canonical Task Contracts

Every task declares an objective, observation contract, feasibility constraints, success condition, and primary metric. Shared observations include pose and relative target cues, local current, and task-specific probes; shared feasibility checks include domain bounds, terrain validity, seafloor clearance, and configured team size.

Go-to-Goal Under Currents (G2G). Reach a goal while rejecting data-grounded drift; success requires entering the goal tolerance, and final goal error is primary.

Station Keeping (SK). Hold position under disturbance; station RMS error measures sustained deviation.

Route Following (RF). Visit an ordered waypoint route; cross-track error and reached waypoints expose partial progress.

Depth-Profile Tracking (DPT). Follow a target depth profile during motion; depth-tracking error is primary.

Area Scan (AS). Cover a bounded region over 3D terrain; grid coverage fraction measures progress.

Pipeline Inspection (PI). Follow a pipeline corridor and encounter leak cues; the inspection score is the detected-leak fraction, with route progress recorded separately.

Formation Transit (FT). Move a team of 2–10 vehicles while maintaining relative geometry; formation error and collision ratio capture coordination quality.

Surface Pollution Cleanup (SPC). Assign 2–10 vehicles to discrete surface sources and satisfy per-source dwell requirements; cleanup rate is completed-source fraction.

Underwater Pollution Lift (UPL). Coordinate five vehicles around a payload and lift it to the surface; lift completion defines success.

Fish Protection and Herding (FH). Use eight vehicles to contain and escort a fish school under currents; centroid progress toward the escort goal defines success.

Exact progress metrics

Several task metrics are normalized ratios with distinct denominators. Let $\mathcal{V}$ be the visited scan cells in an $H \times W$ grid, N_{det} the detected leaks among N_{leak} placed leaks, and N_{done} the sources whose dwell requirement is complete among N_{src} sources. The released evaluator computes

$$\text{Cov}_{\text{AS}} = \frac{|\mathcal{V}|}{HW}, \quad (\text{A5})$$

$$\text{Score}_{\text{PI}} = \frac{N_{\text{det}}}{N_{\text{leak}}}, \quad (\text{A6})$$

$$\text{Rate}_{\text{SPC}} = \frac{N_{\text{done}}}{N_{\text{src}}}, \quad (\text{A7})$$

$$e_{\text{FT}} = \frac{1}{N} \sum_{i=1}^N \|p_i - (p_g + r_i)\|_2, \quad (\text{A8})$$

$$\text{Prog}_{\text{FH}}(t) = \text{clip}\left(1 - \frac{d_t}{d_0}, 0, 1\right). \quad (\text{A9})$$

Here $p_g + r_i$ is agent i 's assigned formation goal, while d_t and d_0 are the current and initial fish-centroid distances to the escort goal. These quantities expose partial progress but are not interchangeable physical measurements.

Tiers 1, 2, and 3 preserve these contracts while jointly changing current gain, tolerances, route or coverage extent, sensing difficulty, and coordination tightness. They are therefore named presets rather than a claim that one scalar fully orders task difficulty. Navigation and tracking tasks primarily stress disturbance rejection, sensing and inspection tasks stress spatial coverage and event encounters, and coordination and pollution tasks add team geometry, assignment, or dwell-time coupling.

State-dependent completion

Navigation and station keeping. Let $d_n = \min_i \|p_i^n - p_g\|_2$. G2G completes when d_n enters the goal radius. SK requires consecutive qualifying steps: its hold counter follows

$$h_{n+1} = \begin{cases} h_n + 1, & d_n \leq r, \\ 0, & d_n > r. \end{cases} \quad (\text{A10})$$

Success occurs when h_{n+1} reaches the prescribed hold length. Consequently, an excursion outside the station radius restarts the hold interval. RF and DPT maintain an ordered waypoint index, advance it when a vehicle enters the active waypoint radius, and evaluate completion from the updated index and that step's waypoint distance. DPT additionally requires the team-mean absolute depth error relative to the evaluated waypoint to satisfy its depth tolerance. FT applies its tolerance to every assigned formation goal: $\max_i \|p_i - (p_g + r_i)\|_2 \leq r$.

Area coverage and inspection. AS stores a persistent binary grid of visited cells. With sensor radius r_s and cell width ℓ , a vehicle marks a square neighborhood extending $\lceil r_s/\ell \rceil$ cells in

each horizontal direction, clipped at the grid boundary. Cells are counted once. A lawnmower route supplies intermediate goals, while completion depends on the coverage fraction in Equation A5. PI maintains a separate detection flag for each leak. A flag becomes permanent when any vehicle enters the leak's detection radius; the first encounter time is recorded. Detection of all leaks completes the task, and the waypoint index records traversal progress separately.

Accumulated service in surface cleanup. For source j , let a_i^n denote agent i 's assigned source and q the required number of agents. Its accumulated service time is

$$b_j^n = \sum_i \mathbf{1}[a_i^n = j] \mathbf{1}[\|p_i^n - s_j\|_2 \leq r], \quad (\text{A11})$$

$$\tau_j^{n+1} = \tau_j^n + \Delta t \mathbf{1}[b_j^n \geq q]. \quad (\text{A12})$$

A source completes when τ_j reaches its dwell threshold. Service already accumulated is retained when vehicles leave; this differs from the consecutive hold in SK. Only vehicles assigned to that source contribute to its service count. The task completes once every source is finished. The hard preset couples assignment with rendezvous by requiring two agents to service a source simultaneously.

Attachment and collective lifting. UPL has four phases: approach, lift-off, fifth-agent joining, and ascent. Each of the five agents acquires a persistent attachment flag upon entering the payload attachment radius. Attachment of the first four agents starts the lift-off hold; completing that hold enables joining, and attachment of the fifth agent enables ascent. During the lifting phases the payload follows the attached team's mean position with a phase-dependent depth update. Completion requires the ascent phase, all five attachments, and payload depth at or above the designated surface threshold.

Fish-school escort. FH evolves a school under current drift and a herding response to nearby vehicles. Its progress uses the horizontal distance between the school centroid and escort goal. The eight-vehicle team succeeds when either the normalized progress in Equation A9 reaches its target or the centroid enters the goal radius.

C Implementation Defaults, Pollution, and ROS 2

Serialized engineering defaults

The main paper defines current coupling, action clipping, diagonal engineering dynamics, and terrain feasibility. Table A4 records the numerical defaults for the six-degree-of-freedom (6DoF) mode serialized in the released run specifications so that those equations can be reproduced without relying on hidden simulator state.

Each run also records whether it uses direct kinematic integration, the diagonal 3DoF mode, or the bounded-attitude 6DoF mode, preventing mode selection from being inferred from output-directory names.

Task	Tier 2 parameters	Tier 3 parameters	Step budget	
			Tier 2	Tier 3
G2G	Goal radius 4 m	Goal radius 1.3 m	280	300
SK	Radius 2.6 m; hold 110 steps	Radius 1.3 m; hold 240 steps	360	560
RF	12 waypoints; radius 3.2 m	20 waypoints; radius 1.6 m	460	560
DPT	12 waypoints; radius 3.6 m; depth tolerance 1.2 m	18 waypoints; radius 1.8 m; depth tolerance 0.7 m	560	720
FT	Formation radius 36 m; goal tolerance 4 m	Formation radius 70 m; goal tolerance 1 m	600	550
SPC	12 sources; radius 3.8 m; dwell 8 s; one agent	6 sources; radius 2.6 m; dwell 6 s; two agents	720	900
UPL	Attach radius 4 m; hold 3.5 s; surface depth 2.5 m	Attach radius 3 m; hold 4 s; surface depth 2.5 m	720	880
FH	220 fish; progress 0.995 or goal radius 12 m	160 fish; progress 0.99 or goal radius 7 m	650	900
AS	Cell 75 m; sensor radius 35 m; coverage 0.82	Cell 60 m; sensor radius 70 m; coverage 0.95	2000	1700
PI	14 waypoints; 6 leaks; detection radius 5 m	20 waypoints; 8 leaks; detection radius 3.6 m	1500	1500

Table A3: Task-specific parameters for the two principal evaluation tiers. Durations follow from the step budget and simulation timestep; the default Δt is 1 s. UPL uses five vehicles and FH uses eight.

Field	Default	Meaning	Field	Default	Meaning
Δt	1.0 s	Outer step	$v_{\max}$	1.2 m s ⁻¹	Relative speed cap
M_ℓ	(12, 12, 12)	Linear inertia proxy	M_a	(6, 6, 6)	Angular inertia proxy
D_ℓ	(8, 8, 8)	Linear damping	D_a	(3, 3, 3)	Angular damping
K_ℓ	(18, 18, 18)	Linear tracking gain	K_a	(10, 10, 10)	Angular tracking gain
K_d	(2, 2, 2)	Angular damping gain	$\omega_{\max}$	(1.2, 1.2, 1.2) rad s ⁻¹	Angular-rate cap
k_{yaw}	2.0	Yaw-rate gain	k_{att}	2.0	Roll and pitch stabilization
$\theta_{\max}$	0.6 rad	Roll and pitch limit			

Table A4: Default 6DoF engineering parameters serialized in each run specification.

Discrete motion update and terrain feasibility

The action a_n specifies desired velocity relative to the water in the world frame. Define the speed projection $\Pi_v(a) = a \min(1, v_{\max}/\|a\|_2)$ for nonzero a and $\Pi_v(0) = 0$. In kinematic mode the candidate position is

$$\tilde{p}_{n+1} = p_n + \Delta t[\Pi_v(a_n) + c_n], \quad (\text{A13})$$

where c_n is the sampled world-frame current, including the configured disturbance terms.

For the dynamic modes, let R_n map body vectors to the world frame, $\nu_n = (v_n, \omega_n)$ be body-frame linear and angular velocity, and ω_n^{cmd} the desired angular rate. The latter turns yaw toward the horizontal action direction and stabilizes roll and pitch toward zero. The force and moment commands are

$$f_n = K_\ell[R_n^\top \Pi_v(a_n) - v_n], \quad (\text{A14})$$

$$m_n = K_a(\omega_n^{\text{cmd}} - \omega_n) - K_d \omega_n. \quad (\text{A15})$$

With the diagonal parameters in Table A4, the velocity update is

$$\tilde{v}_{n+1} = v_n + \Delta t M_\ell^{-1}(f_n - D_\ell v_n), \quad (\text{A16})$$

$$\tilde{\omega}_{n+1} = \omega_n + \Delta t M_a^{-1}(m_n - D_a \omega_n). \quad (\text{A17})$$

Linear speed is projected with Π_v and angular rates are clipped componentwise before advancing position:

$$\tilde{p}_{n+1} = p_n + \Delta t[R_n \tilde{v}_{n+1} + c_n]. \quad (\text{A18})$$

Attitude advances with the clipped angular rates; yaw is wrapped to its principal interval and roll and pitch are bounded by $\theta_{\max}$. The 3DoF mode holds roll and pitch, and their rates, at zero. Position uses the rotation at the start of the step and the updated body velocity.

The candidate position is first clamped to the configured spatial bounds. At this candidate location, valid water requires

an admissible mask cell and depth no greater than seabed depth minus the clearance margin, whose default is 1 m. A terrain-infeasible candidate retains the previous position and increments the constraint-violation count; in dynamic modes it also resets relative linear velocity to zero. Near-collision events are measured from pairwise distances after motion. They contribute to the coordination metric without altering the motion update. Task state and completion are then evaluated on the resulting vehicle positions.

Pollution fields and cleanup semantics

The analytic probe field is a current-advected Gaussian,

$$C(p, t) = m(t) C_{\text{peak}} \exp[-\|p - \mu(t)\|^2 / (2\sigma^2)], \quad (\text{A19})$$

with default $\sigma = 35$ m and $C_{\text{peak}} = 1$. An agent in the 35 m continuous-containment neighborhood applies $m(t + \Delta t) = m(t) \max(0, 1 - 0.12\Delta t)$.

Three-dimensional transport and transformation model

The Ocean Current Prediction Network (OCPNet) pollution engine evolves each configured species s through the implemented balance

$$\begin{aligned} \frac{\partial C_s}{\partial t} = & -\mathbf{u} \cdot \nabla C_s + \nabla \cdot (K_s \nabla C_s) - \lambda_s C_s \\ & + \sum_r \nu_{sr} R_r + U_s + \mathcal{P}_s + S_s - Q_s, \end{aligned} \quad (\text{A20})$$

$$\begin{aligned} R_r = & k_{r0} \exp[\alpha_r (T - 298.15)] f_r(\text{pH}) \\ & \times \prod_{j \in \mathcal{R}_r} C_j, \end{aligned} \quad (\text{A21})$$

$$f_r(\text{pH}) = \max(1 + a_r \Delta \text{pH} + b_r \Delta \text{pH}^2, 0).$$

Here $\Delta \text{pH} = \text{pH} - 7$, C_s is concentration, $\mathbf{u} = (u, v, w)$ is velocity, K_s is diffusivity, λ_s is first-order decay, and ν_{sr} is a

signed stoichiometric coefficient. The optional terms $\mathcal{U}_s$ and $\mathcal{P}_s$ represent biological uptake and precipitation, while S_s and Q_s collect externally configured sources and sinks.

The code computes spatially varying diffusivity from temperature T , wave velocity U_w , and salinity S as

$$\begin{aligned} K_T &= K_{s0} [1 + 0.02(T - 273.15)], \\ K_w &= K_T + 0.1U_w, \\ K_s &= K_w [1 + 0.05(1 - S/35)]. \end{aligned} \quad (\text{A22})$$

These coefficients reproduce the released numerical parameterization rather than a calibrated turbulence closure. When decay is enabled, its implemented environmental dependence is

$$\begin{aligned} \lambda_s &= \lambda_{s0} \exp\left(-\frac{E_{a,s}}{RT}\right) f_s(\text{pH}) f_s(O), \\ f_s(\text{pH}) &= \max(1 + a_s \Delta \text{pH} + b_s \Delta \text{pH}^2, 0), \\ f_s(O) &= \max(1 + c_s O + d_s O^2, 0), \end{aligned} \quad (\text{A23})$$

where the exponential factor is omitted when $E_{a,s} = 0$, R is the gas constant, and O is dissolved oxygen. Biological uptake, when configured, contributes the Michaelis–Menten sink

$$\mathcal{U}_s = -V_{\max,s} e^{\alpha_{T,s}(T-298.15)} (1 + \alpha_{L,s} L) \frac{C_s}{K_{1/2,s} + C_s} B, \quad (\text{A24})$$

with light L and phytoplankton biomass B . For a configured cation–anion precipitation pair, the saturation and rate terms are

$$\begin{aligned} \text{SI} &= \frac{C_M C_A}{K_{\text{sp}}}, \\ r_p &= k_p e^{\alpha_p(T-298.15)} f_p(\text{pH}) \max(\text{SI} - 1, 0), \end{aligned} \quad (\text{A25})$$

which subtract r_p from both dissolved species and add it to the configured precipitate before its vertical settling term is applied.

For transport, the implementation uses a first-order upwind difference. Along x , for example,

$$(\delta_x^{\text{up}} C)_i = \begin{cases} (C_i - C_{i-1})/\Delta x, & u_i \geq 0, \\ (C_{i+1} - C_i)/\Delta x, & u_i < 0, \end{cases} \quad (\text{A26})$$

and the centered flux form of the corresponding diffusion term is

$$(\mathcal{D}_x C)_i = \frac{K_i(C_{i+1} - C_i) - K_{i-1}(C_i - C_{i-1})}{\Delta x^2}. \quad (\text{A27})$$

The same constructions are applied along y and z , and explicit Euler combines every active process:

$$\begin{aligned} C_s^{n+1} &= C_s^n + \Delta t \mathcal{F}_s^n, \\ \mathcal{F}_s &= \mathcal{A}_s + \mathcal{D}_s - \lambda_s C_s + \sum_r \nu_{sr} R_r \\ &\quad + \mathcal{U}_s + \mathcal{P}_s + S_s - Q_s. \end{aligned} \quad (\text{A28})$$

The solver rejects velocity fields that violate

$$\max\left(\frac{\|u\|_\infty \Delta t}{\Delta x}, \frac{\|v\|_\infty \Delta t}{\Delta y}, \frac{\|w\|_\infty \Delta t}{\Delta z}\right) \leq 1, \quad (\text{A29})$$

and the diffusion module exposes the conservative helper bound $\Delta t_{\text{diff}} = \frac{1}{2} \min(\Delta x, \Delta y, \Delta z)^2 / \|K_s\|_\infty$. Dirichlet, Neumann, periodic, and open boundaries are configurable. The general three-dimensional update does not clip negative concentrations, so stable timesteps and physically consistent process parameters remain the user’s responsibility.

The benchmark wrapper uses a $28 \times 28 \times 10$ grid, a 2 s internal step, base diffusivity 8×10^{-8} , point-source emission rate 0.02, 8 m sink radius, and sink strength 0.15 s^{-1} . Its default benchmark case activates transport, diffusion, the point source, and the robot sink; the other transformation terms above are modular capabilities and are not silently activated. Dataset horizontal current is resampled and repeated over model depth with zero vertical current. This is an explicit approximation, not a measured three-dimensional velocity profile. The robot sink multiplies cells in its local grid neighborhood by $\max(0, 1 - 0.15 \Delta t)$ at each internal update.

Dataset-driven surface multispecies proxy

The supplementary visualization uses a separate two-dimensional proxy driven by a selected depth slice of the dataset currents. For any field F , one transport substep is

$$\mathcal{T}_n(F) = F + \Delta t [-u_n F_x - v_n F_y + \kappa(F_{xx} + F_{yy})]. \quad (\text{A30})$$

Let A , B , and C denote normalized microplastic, crude-oil, and aggregate fields, respectively. Transport first gives $A^* = \mathcal{T}_n(A^n)$, $B^* = \mathcal{T}_n(B^n)$, and $C^* = \mathcal{T}_n(C^n)$. With $[x]_+ = \max(x, 0)$, aggregation and weathering are then

$$r_a = k_a [A^*]_+ [B^*]_+, \quad r_w = k_w [B^*]_+, \quad (\text{A31})$$

and the implemented update before masking and normalization is

$$\begin{aligned} \tilde{A} &= [A^* - \Delta t r_a]_+ + 0.010 S_A, \\ \tilde{B} &= [B^* - \Delta t(r_a + r_w)]_+ + 0.008 S_B, \\ \tilde{C} &= [C^* + \Delta t r_a]_+. \end{aligned} \quad (\text{A32})$$

Gaussian source maps S_A and S_B continuously replenish the two parent fields. Land-mask cells are then set to zero, and every nonzero species field is independently divided by its spatial maximum. Consequently, Figure A2 shows relative plume shape and reaction overlap rather than calibrated concentration or mass-conserving forecasts. This surface proxy uses centered numerical gradients and must not be conflated with the upwind three-dimensional solver in Equations A20–A29.

Building on the distinct localization, containment, and discrete-cleanup semantics defined in the main paper, the exact canonical SPC presets are: 12 sources, 8 s dwell, and one required agent at medium difficulty; 6 sources, 6 s dwell, and two required agents at hard difficulty.

Optional Robot Operating System 2 (ROS 2) Jazzy bridge

The bridge maps core $(x, y, z) = (\text{east}, \text{down}, \text{north})$ to the ROS east-north-up (ENU) frame $(x, z, -y)$. External commands traverse the same clipping, dynamics, current, constraints,

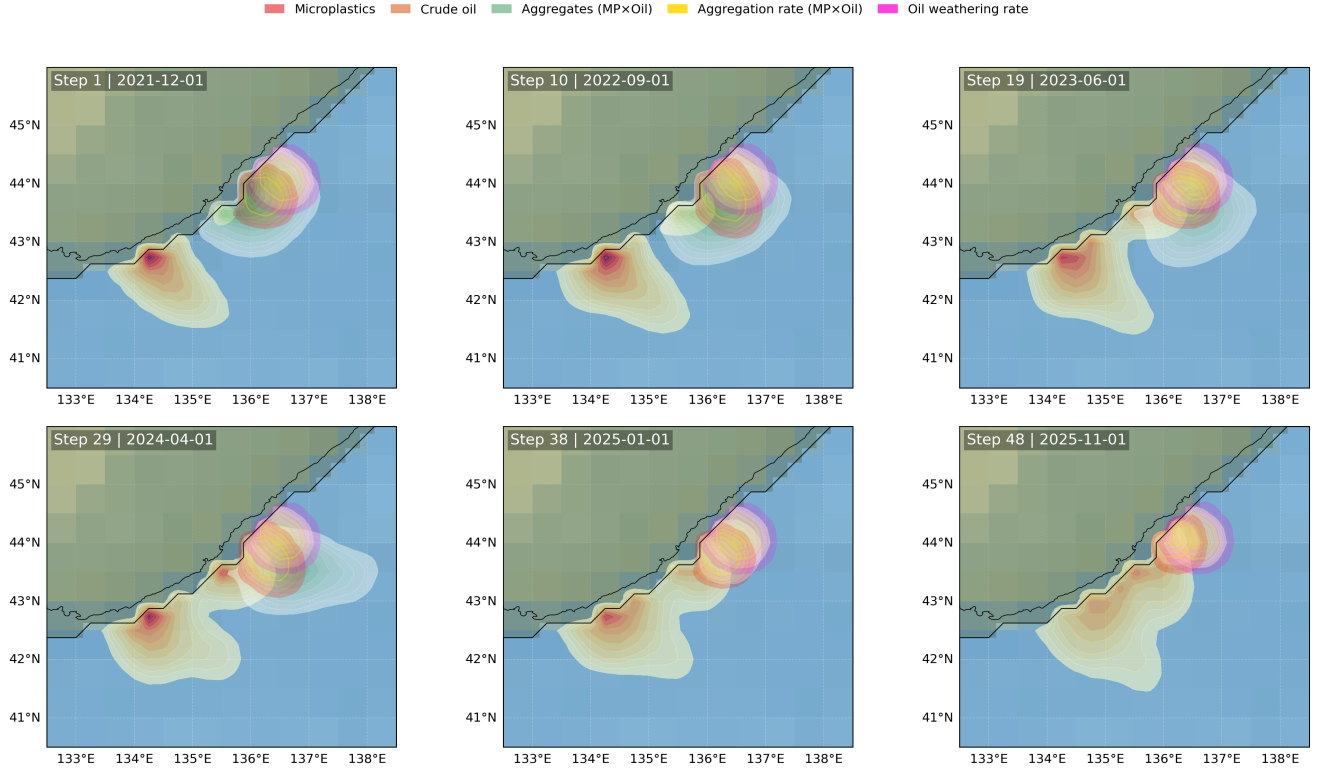


Figure A2: Six sampled times from the dataset-driven surface multispecies proxy. The overlays show normalized microplastic, crude-oil, and aggregate fields together with aggregation and oil-weathering rates. The land mask blocks the proxy fields. Because each species is normalized independently after every substep, this panel is qualitative evidence of transport and reaction structure rather than a mass forecast.

Interface	ROS type and meaning
/agent_NNN/odom	nav_msgs/Odometry; ENU pose and velocity
/agent_NNN/current	Vector3Stamped; ENU current
/agent_NNN/pollution	Float32; local scalar probe
/agent_NNN/bathymetry	Float32; positive water depth
/agent_NNN/cmd_vel	subscribed ENU velocity command
/clock, /episode/metrics	simulation time and JavaScript Object Notation episode state
/reset, /step, /get_task_state	std_srvs/Trigger services

Table A5: ROS 2 interface. Topic prefixes are /oneocean/agents; agent IDs are zero-padded.

recorder, and task-update path as built-in controllers; internal-control mode can publish observations without accepting external motion commands.

The package was built and launched in a ROS 2 Jazzy container; topic discovery, one externally commanded step, state query, and reset completed successfully. This is an interface smoke test, not a ROS performance result, and ROS remains an optional dependency outside the core requirements.

D Metrics and Reproducibility Protocol

Metric definitions

The main paper defines success rate (SR), timeout-aware episode time T , and the action-energy proxy E . The additional team-level aggregate used in the result tables is the near-collision ratio

$$\text{Coll} = \frac{1}{T_{\text{ep}}} \sum_{t=1}^{T_{\text{ep}}} \mathbf{1} \left[\min_{i < j} \|p_t^{(i)} - p_t^{(j)}\|_2 \leq r_{\text{coll}} \right]. \quad (\text{A33})$$

Task-primary metrics follow the per-task contracts in Appendix Sec. B and are not pooled across incompatible physical meanings.

Episode and task aggregation

For task k with M_k episodes, let $s_{ke} \in \{0, 1\}$ indicate success and H_{ke} the timeout in seconds. The reported success rate and timeout-aware duration are

$$\text{SR}_k = \frac{1}{M_k} \sum_e s_{ke}, \quad (\text{A34})$$

$$T_k = \frac{1}{M_k} \sum_e [s_{ke} t_{ke}^{\text{succ}} + (1 - s_{ke}) H_{ke}], \quad (\text{A35})$$

where t_{ke}^{succ} is the success time for a completed episode. The success-only duration, when reported separately, averages t^{succ} over successful episodes and is undefined if none succeeds.

An episode's action-energy proxy sums over both agents and executed steps:

$$E_{ke} = \Delta t \sum_{n=0}^{L_{ke}-1} \sum_{i=1}^N \|\Pi_v(a_{i,n})\|_2^2. \quad (\text{A36})$$

Here L_{ke} is the executed step count. The recorded current and realized velocity do not enter this command-based quantity. The near-collision ratio above uses one indicator per step, even if several agent pairs are close in that step; tables expressed in percent multiply this ratio by 100.

Station error is accumulated after each motion update. Writing $\bar{d}_n = N^{-1} \sum_i \|p_i^n - p_g\|_2$, its episode RMS is

Diagnostic	Budget	Configuration	Purpose
Planner audit	3 seeds per task and method	72 episodes; stride 200; 96-token cap	Validity, fallback, completion, and failure diagnosis.
Dynamics diagnostic	20 episodes per task and difficulty	10 tasks; medium and hard; three dynamics modes	Sensitivity to engineering dynamics.
Current-gain sweep	5 seeds per task and method	300 episodes; hard; 10 tasks; $g \in \{0, 1, 2\}$	Data-current disturbance sensitivity.
Synthetic-tide stress	10 seeds per task and method	200 episodes; hard; 10 tasks; 0.45 m s^{-1} amplitude	Periodic-disturbance sensitivity.
External pollution diagnostic	3 episodes per scene and configuration	Four scenes; two currents; two field models	Localization-model comparison.
External scaling	10 episodes per scene and team size	80 episodes; two scenes; $N \in \{2, 4, 8, 10\}$	Scene-backed team-size sensitivity.

Table A6: Budgets for supplementary diagnostic studies. Headline experiment budgets remain attached to their result tables in the main paper.

$[L^{-1} \sum_n \bar{d}_n^2]^{1/2}$. The route-tracking diagnostic analogously averages distance to the active waypoint across agents and time. It measures the waypoint-target error associated with the current route stage. For an overall success rate across K tasks, the macro-average is $K^{-1} \sum_k \text{SR}_k$, which gives every task equal weight. With equal episode counts this also equals the pooled episode success rate. Physical task metrics retain their own units and are averaged within their task and configuration.

Controller observations and decisions

The heuristic, BC, and LLM-assisted controllers share the action interpretation and the motion update in Appendix Sec. C. Heuristics track task targets, BC maps observations to velocity, and the LLM selects discrete task decisions.

Goal-directed control. For an assigned target q_i , the proportional controller produces

$$a_i = \Pi_v(k_p(q_i - p_i)). \quad (\text{A37})$$

The navigation gain is $k_p = 0.9$ and the station-keeping gain is 0.8 . Intermediate targets come from the task state: the active waypoint, assigned source, formation offset, or stage of a coordinated task. This controller corrects positional error at each step under the applied current. The pollution-localization heuristic additionally uses the agents’ scalar probes to estimate a horizontal concentration gradient and coordinate exploration around the highest observed concentration.

Behavior cloning. The observation for each training example is

$$o_i = [(q_i - p_i)^\top, y_i, C(p_i), c_{i,x}, c_{i,z}, e_k^\top]^\top, \quad (\text{A38})$$

where e_k is a one-hot task indicator. It contains seven continuous features followed by the task encoding. The policy is a multilayer perceptron with two ReLU hidden layers and a three-dimensional linear output. Feature and action means and standard deviations are estimated on the training split. With $\hat{o}$ and $\hat{a}$ denoting standardized values, the learning objective is

$$\mathcal{L}_{\text{BC}}(\theta) = \frac{1}{3M} \sum_{m=1}^M \|f_\theta(\hat{o}_m) - \hat{a}_m\|_2^2. \quad (\text{A39})$$

AdamW minimizes this objective. At inference the output is transformed back to action units and projected onto the permitted speed ball. The reference training configuration uses hidden widths (64, 64), 12 epochs, batch size 4096, learning rate 10^{-3} , zero weight decay, and a randomly selected 10% validation split; the trained checkpoint records its configuration and normalization statistics.

Demonstrations align the recorded command with its task target and local observations. The trajectory converter retains steps with a recorded target. For the kinematic recording convention, which stores the post-step position, it reconstructs the corresponding pre-step position as $p_n = p_{n+1} - \Delta t(a_n + c_n)$ before forming the target displacement.

LLM-assisted planning. At a planning event, the task state is summarized by the task identifier, step, team size, agent positions, and the relevant source or waypoint state. Coordinates are rounded to two decimal places. The planner returns discrete source assignments or waypoint choices, which are checked for admissible indices and task state before use. The task controller then tracks the resulting targets using the same bounded velocity control. Between planning events the current decisions remain active. Failed or invalid responses invoke the task’s deterministic fallback. The audit in Appendix Sec. E separates decision validity, fallback frequency, task completion, and inference cost, to characterize discrete planning within the closed-loop episode.

Supplementary diagnostic budgets

All diagnostics reuse the same serialized tier presets, controller settings, dataset indices, and environment provenance as their corresponding headline study; only the diagnostic factors named in Table A6 are varied.

Compute and run manifests

The main paper reports the compute platform. For provenance, each run stores an integer seed, normalized configuration, task and dynamics specification snapshots, environment-product indices, software and Git state, episode metrics, and recorder streams. Aggregation scripts consume these artifacts to regenerate comma-separated-value and Markdown tables rather than transcribing console output.

The reproducibility boundary is explicit: data validation and ROS adapter tests are release tests; they are not added to performance averages. Likewise, the three-seed planner audit and three-episode-per-scene pollution comparison diagnose mechanisms but do not replace the larger headline budgets.

Model boundaries and extension priorities

The pollution fields are controlled sensing signals rather than in-situ labels or validated spill forecasts. Surface Pollution Cleanup completes discrete sources after a dwell requirement, whereas continuous containment removes mass from a concentration field. Underwater Pollution Lift instead measures coordinated payload lifting; these objectives have distinct completion criteria. Tide attenuation remains an engineering

Hard task	Kinematic SR	3DoF SR	6DoF SR
Formation Transit	55%	50%	50%
Surface Cleanup	95%	85%	85%
Pollution Lift	100%	75%	75%
All-task macro	85.5%	81.5%	81.5%

Table A7: Selected dynamics sensitivity at hard difficulty (20 episodes per task and mode).

Stress	Method	<i>N</i>	SR	<i>E</i>	Coll.
Data current, $g = 0$	Heuristic	50	48.0	3232.5	5.80
Data current, $g = 1$	Heuristic	50	20.0	3371.6	8.24
Data current, $g = 2$	Heuristic	50	8.0	3775.0	9.86
Data current, $g = 0$	BC	50	6.0	1546.8	4.14
Data current, $g = 1$	BC	50	12.0	1807.1	4.96
Data current, $g = 2$	BC	50	4.0	2323.3	2.44
Synthetic tide	Heuristic	100	17.0	3489.2	6.29
Synthetic tide	BC	100	1.0	1557.6	6.49

Table A8: Archived hard-setting disturbance diagnostics across all ten tasks. SR and Coll. are percentages. Each data-current row contains five episodes per task. The synthetic-tide rows contain ten episodes per task, use $g = 0$, and apply a 0.45 m s^{-1} sinusoid with a 300 s period. These focused sweeps diagnose sensitivity and are separate from the headline budgets.

approximation rather than a phase-accurate three-dimensional tide model. The ROS 2 bridge has passed message, reset, frame-conversion, and command smoke tests, but it has not been benchmarked for latency, determinism, middleware load, or hardware-in-the-loop performance. The highest-priority extensions are measured three-dimensional flow, identified vehicle coupling through computational fluid dynamics (CFD) or system identification, physically validated pollutant transport, and ROS-in-the-loop experiments. These additions can preserve the current task and reporting contracts while increasing physical realism.

E Additional Core-Backend Diagnostics

Dynamics-mode sensitivity

Table A7 selects the hard-setting tasks with the largest observed dynamics sensitivity from a 20-episode-per-task diagnostic.

Across all ten hard tasks, macro SR is 85.5% for kinematic and 81.5% for both 3DoF and 6DoF. The identical 3DoF and 6DoF SRs under this controller do not establish physical equivalence; they show that bounded roll and pitch did not change terminal success in this diagnostic, whereas replacing the velocity state with direct kinematic integration did.

Current and synthetic-tide stress

Heuristic SR decreases from 48.0% without data-current forcing to 8.0% at doubled current, while its energy and collision exposure increase. BC is not monotonic at the smaller gains, but reaches its lowest SR at $g = 2$. The synthetic-tide stress also separates the methods, with 17.0% SR for the heuristic and 1.0% for BC. Because the current-gain cells contain only five episodes per task, these results establish a sensitivity trend rather than precise task-level rankings.

Planner validity, fallback, and cost

The instrumented audit contains three medium-difficulty seeds for each of eight methods and three tasks, for 72 episodes total. Tables A9 and A10 aggregate each large language model (LLM) across its nine episodes. Valid-plan ratio is weighted by planner calls; latency and tokens are weighted by uncached calls.

LLM	SR	Calls	Valid	Fallback	Comp.
ChatGLM3-6B	0.556	49	0.367	0.633	0.758
Llama2-7B	0.333	60	0.950	0.050	0.736
Llama3-8B	0.444	61	0.869	0.131	0.786
Mistral-7B	0.556	55	0.800	0.200	0.804
Qwen2-7B	0.556	43	0.814	0.186	0.790
Qwen2.5-7B	0.667	59	0.864	0.136	0.809

Table A9: Per-model planning outcomes across SPC, AS, and PI (nine episodes per LLM).

LLM	Latency ms per call	Prompt tokens per call	Output tokens per call
ChatGLM3-6B	1466.1	628.1	76.7
Llama2-7B	7429.6	633.7	728.7
Llama3-8B	2068.1	454.1	475.9
Mistral-7B	5150.6	616.5	676.7
Qwen2-7B	2370.6	603.3	628.6
Qwen2.5-7B	2184.5	600.6	623.9

Table A10: Uncached per-call planning cost in the same audit.

Model-level and task-level rankings also diverge: Llama2-7B attains 0.950 validity but only 0.333 SR, whereas ChatGLM3-6B attains 0.367 validity and 0.556 SR. At task level, cleanup’s low valid ratio coexists with high completion because invalid or duplicate assignments can fall back to a competent deterministic allocator; AS can revisit covered cells despite valid syntax, and PI still requires physical leak encounters between low-frequency plans. Planner validity is therefore a mechanism diagnostic, not a substitute for task success.

F Additional External-Backend Diagnostics

The external pollution study compares an analytic advected Gaussian probe with OCPNet-3D in Dam, OpenWater, Pier-Harbor, and SimpleUnderwater. Each combination of scene, model, and current setting uses three localization episodes; Table A12 therefore aggregates 12 episodes per row. It is a focused observation-model diagnostic rather than a high-confidence performance ranking.

OCPNet-3D lowers mean localization error by 5.12 m at nominal current and 3.36 m under stronger current, while matching or improving aggregate success. This result supports the narrower claim that richer regeneratable concentration structure can change sensing difficulty; it does not validate OCPNet as a real spill forecast. Containment outputs are not collapsed into the same table because analytic remaining-mass state and OCPNet grid mass have different scales and should not be compared as a shared physical quantity.

Scene-backed cleanup scaling

Increasing the team from two to four agents raises aggregate success from 40% to 100% and approximately doubles removed

Task	SR	Valid	Fallback	Latency ms per call	Completion
SPC	0.833	0.430	0.570	3199.0	0.972
AS	0.278	0.831	0.169	3602.0	0.758
PI	0.444	0.891	0.109	3337.1	0.611

Table A11: Cross-model planner diagnostics by task (six LLMs, three seeds each).

Scale	Probe	N	SR (%)	Error (m)	Δ error (m)
1.0	Analytic	12	41.7	25.93	0.00
1.0	OCPNet-3D	12	50.0	20.81	-5.12
2.5	Analytic	12	50.0	26.00	0.00
2.5	OCPNet-3D	12	50.0	22.64	-3.36

Table A12: Pollution-localization diagnostic over four scenes (12 episodes per row). Δ error is measured against the analytic probe at the same current scale.

Agents	Episodes	SR (%)	Removed mass (kg)
2	20	40.0	0.2081
4	20	100.0	0.4302
8	20	100.0	0.4817
10	20	100.0	0.4886

Table A13: Analytic-containment scaling at nominal current, aggregated over OpenWater and PierHarbor with ten episodes per scene and team size.

mass. The smaller gains from four to ten agents indicate saturation under the fixed 15 s containment horizon. Removed mass is comparable within this analytic-field sweep only and is not equated with OCPNet grid mass.

The route-transfer results retained in the main paper reveal substantial scene dependence despite a shared task contract. Together, the route and pollution diagnostics motivate reporting scene, current scale, dataset clock, controller profile, and observation model in every external-backend manifest rather than treating “external simulator” as a single condition.